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Paleomagnetic study of French Guyana Early Jurassic dolerites: hypothesis of a multistage magmatic event

S. Nomade [a,*], H. Théveniaut [b], Y. Chen [a], A. Pouclet [a], C. Rigollet [a]

[a] *Département des Sciences de la Terre, UMR 6530, ISTO, Université d'Orléans, 45067 Orléans Cedex 2, France*
[b] *BRGM, SGN, 3, avenue Claude Guillemin, P.O. Box 6009, 45060 Orléans Cedex 2, France*

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Abstract

A detailed paleomagnetic and anisotropy of magnetic susceptibility (AMS) study was carried out on 34 sites of Early Jurassic dolerite dykes from French Guyana, which formed during the initial opening of the Central Atlantic Ocean. Four types of AMS fabrics are recognized: (i) 'Normal' fabric (21 dykes) defined by clustering of K_1–K_2 axes on the dyke plane whereas the K_3 axis is nearly perpendicular to it. This fabric is interpreted as due to magma flow. The subhorizontal inclination of the K_1 axis permitted to suggest that the French Guyana dykes could be fed by horizontal magma fluxes from a distant magma source. (ii) 'Reversal' fabric (8 dykes) is characterized by the K_2–K_3 plane close to the dyke plane and the K_1 perpendicular to dyke orientation. Such fabric was attributed to the local shearing stress. (iii) 'Intermediate' fabric (1 dyke) is defined by K_1–K_3 axes close to the dyke plane and K_2 axis is perpendicular to this plane. It was interpreted as due to vertical compaction of a static magma column. (iiii) 'Other' fabric (4 dykes) does not show any preferential orientation. Scanning electronic microscope and susceptibility versus temperature experiments show that minerals of the titanomagnetite family are main magnetic remanence carriers. Two magnetic components were isolated. Ages of magnetic remanences are estimated at 198.3 ± 2.0 Ma to 192.3 ± 1.5 Ma. Their virtual geomagnetic poles are calculated, Pole A: $\lambda = 73.2°$N, $\phi = 15.3°$E, $k = 288.8$, $A_{95} = 3.4°$, $n = 8$, and Pole B: $\lambda = 81.6°$N, $\phi = 89.1°$E, $k = 69.8$, $A_{95} = 4.2°$, $n = 18$. These two groups probably correspond to two distinct magmatic events which occurred in a short period. This hypothesis is consistent with published ^{40}Ar–^{39}Ar radiometric ages though with 'mini plateau' spectra. These paleomagnetic results suggest the presence of magmatic pulses led to the construction of the Central Atlantic Magmatic Province in French Guyana during the Early Jurassic. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; anisotropy; magnetic susceptibility; French Guiana; diabase; magmatism

1. Introduction

Extensive dyke swarms cut both Paleoproterozoic basement rocks of the Guyana shield and sedimentary rocks of the Amazonian basin. They are mostly present along the coastal margins, however inland large basaltic floods associated with the dykes are also widespread. These swarms, such as the Cassipore and Cayenne swarms in Amapa (Brazil) and French Guyana are dominantly NNW–SSE-oriented, nearly paral-

* Corresponding author: Fax: 33-2-38-41-73-09;
E-mail: sebastien.nomade@univ-orleans.fr

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lel to the rifting direction of the Central Atlantic Ocean [1] (Fig. 1). Similar dyking events have been recognized around the Central Atlantic Ocean, from the European and North American continental margins, to the West African and South American coasts, thus constituting the largest volcanic province in the world: the Central Atlantic Magmatic Province (CAMP) [2]. ^{40}Ar–^{39}Ar and ^{238}U–^{206}Pb ages [2,3] of the CAMP basaltic magmatism cluster around 200 Ma, close to the Triassic–Jurassic boundary. If this magmatic event is commonly associated with the initiation of continental rifting as suggested by several workers [4,5], the migration of magma from partial melting zones in the mantle towards the surface is poorly understood. The magma could be injected directly from the mantle source [6] or laterally injected from shallow magma chambers fed by vertical conduits connected to the mantle [7]. The understanding of timing, emplacement, tectonic control and identified magma sources could allow to answer these questions. In particular, the anisotropy of magnetic susceptibility technique (AMS) offers a powerful tool to inves-

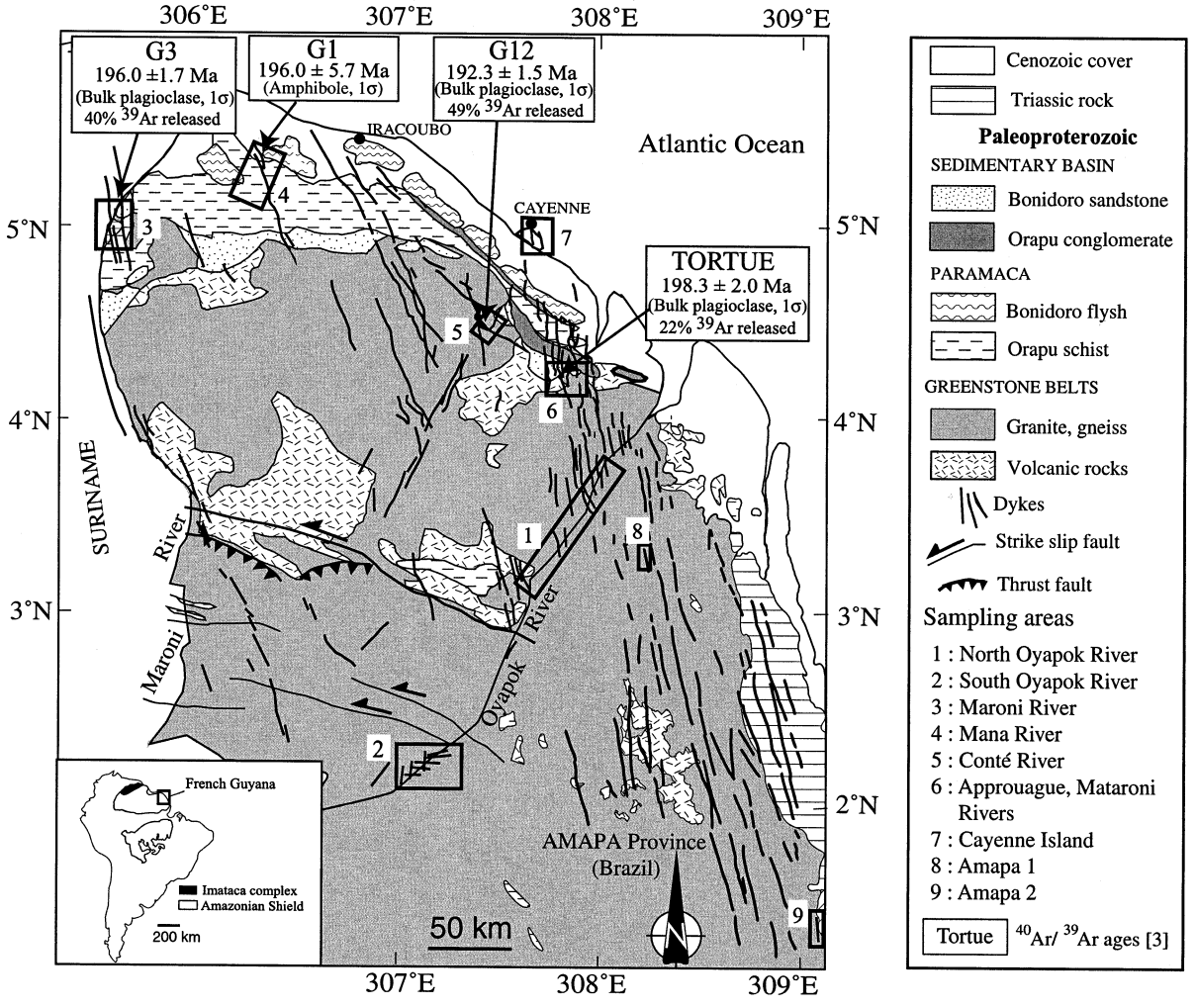


Fig. 1. Simplified geological map of French Guyana showing the South American Amazonian Shield, location of the sampling areas and recently dated dykes [3].

tigate flow directions, proxy flow indicator, or stress direction [8–10]. Moreover, coupling radiochronological ages with paleomagnetic results should better constrain the predrift magmatic process and timing of dyke emplacement. Up to now, no AMS study has been carried out on Early Jurassic dykes in South America. Until recently, paleomagnetic data concerning Early Jurassic dolerites of the Amazonian Shield were not satisfactory enough as they were scarce and only supported by ^{40}K–^{40}Ar dating [11–14]. Recent geochronological ^{40}Ar–^{39}Ar results on the basaltic dykes from French Guyana [3] are also helpful for the understanding of the magmatism of the CAMP in French Guyana.

In the framework of a multidisciplinary BRGM (French Geological Survey) geological mapping project of the French Guyana Territory in collaboration with ISTO and the CPRM (Geological Survey of Brazil), we carried out four field trips between 1995 and 1998. We collected samples from all representative formations and a particular care was taken on dykes, some being recently dated [3,15]. Indeed, a better understanding of dyke emplacement in age and space should better constrain the predrift fracturation process which affected French Guyana during Early Jurassic.

2. Geological setting and paleomagnetic sampling

2.1. Geological setting

The Guyana Shield, which underlies French Guyana is mainly composed of granite–greenstone belts ranging in age from 2.2 to 2.0 Ga due to the Transamazonian tectonothermal event [16] (Fig. 1), with evidence of Archean basement on its western margin (Imataca Complex, Venezuela, [17]). In French Guyana, the basement mainly consists of plutonic rocks and greenstone belts forming two large zones intruded by large granitic complexes [18] (Fig. 1). The northern part of Guyana is occupied by an EW-elongated Paleoproterozoic sedimentary basin filled by the Orapu conglomerate and the Bonidoro sandstone Formations [18,19] (Fig. 1).

Two principal doleritic magmatic events have affected the Guyana Shield: the Paleoproterozoic Avanavero Intrusive Group and the Early Jurassic Apatao Group [11]. In French Guyana the majority of the dyke-swarms are NNW–SSE-trending Early Jurassic basalt [3] (Fig. 1). All investigated rocks have basaltic (sensu lato) compositions [20] (Fig. 2) with intergranulare, ophitic, (sub)ophitic or gabbroic textures. The (sub)-ophitic texture is observed in the majority of dykes. The mineralogy consists of olivine (Fo_{40}–Fo_{70}), plagioclase (An_{40}–An_{60}) and augite. The groundmass is composed of fine-grained plagioclase, augite ± pigeonite, skeleton-like Fe–Ti oxides, and, possibly biotite and/or hornblende. French Guyana dolerites are characterized by a lack of LILE enrichment (Ba, Rb, Th), primitive mantle normalized (La/Yb)n ratios ranging from 1.8 to 4.6, and high TiO_2 ($< 2\%$). They likely correspond to a mixed MORB–EMII mantle source [15].

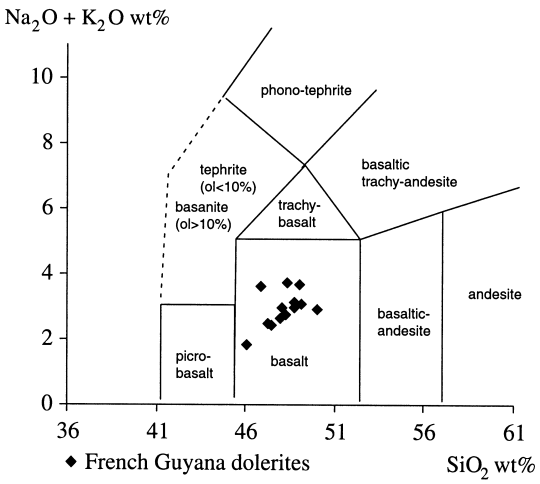


Fig. 2. Total alkalis versus silica diagram (T.A.S) [20].

2.2. Paleomagnetic sampling

In French Guyana, Jurassic dykes are numerous, well exposed, and cut vertically through Paleoproterozoic rocks. Their width varies from 1 to 30 m but generally does not exceed 3–5 m. Samples were collected from 32 sites along the Oyapok, Maroni, Mana, Conté, Approuague and Mataroni rivers, and on the Cayenne Island

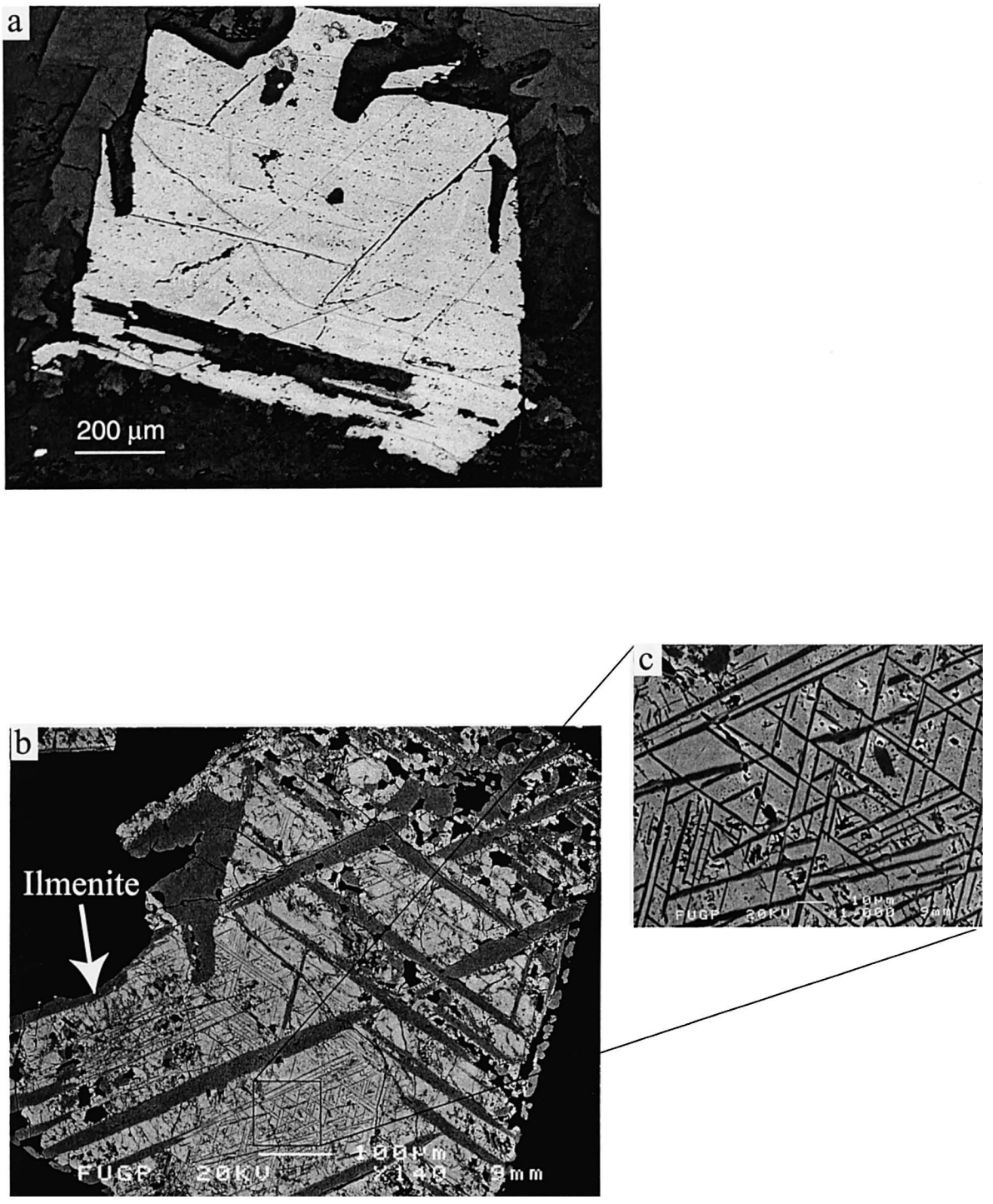


Fig. 3. (a) Reflected light photo of a large automorphous titanomagnetite grain. (b) Large-scale image in secondary electrons of titanomagnetite with the complex structure of ilmenite exsolution. (c) Close scale image in secondary electrons of fractal disposition of ilmenite exsolution in titanomagnetite.

(Fig. 1). Two dykes were also sampled in Amapa (Fig. 1). Four sites were recently dated by the ^{40}Ar–^{39}Ar method [3,15] (Fig. 1). Dyke azimuths vary from N330 to N020 and were measured in 21 dyke walls. The other azimuths (13 dykes) were interpreted from the geophysical data. In all well exposed dykes (21), a fine congealed margin (<2 cm) was observed. Six to eight 2.54 cm in diameter cylinders crossing through the dyke in width were drilled perpendicular to the strike of the dyke at each site. Cores were orientated with magnetic and, whenever possible, with sun compass. The difference between solar and magnetic azimuth measurements is about $20\pm4°$, which is compatible with the 1998 IGRF declination (17.5°). Samples were also taken for mineralogical and chemical analyses. Cores were cut to standard size (2.54/2.2 cm), yielding 260 specimens.

3. Laboratory measurements

3.1. Petrographic and magnetic mineralogical analyses

In order to characterize the magnetic mineralogical compositions of our collection, we applied the following methods on representative samples: reflection microscopy (Olympus BX60) at the geological laboratory of the Université d'Orléans; electronic scanning microscopy (JEOL instrument) at Ecole Supérieure de l'Energie et des Matériaux (ESEM), and thermomagnetic experiments using a CS3 apparatus (AGICO, Geofysica) coupled with a KLY-3S kappabridge (AGICO, Geofysica) at the joint BRGM/Université d'Orléans Laboratoire de Magnétisme des Roches (LMR).

Table 1
Titanomagnetite microprobe analyses showing a variable concentration of titanium oxides

	GR2	FU8	GW1	HA6
SiO_2	0.05	0.10	0.11	0.17
TiO_2	4.80	8.23	19.44	11.23
Al_2O_3	2.96	1.18	2.26	1.93
Cr_2O_3	0.00	0.26	0.20	0.10
V_2O_3	0.96	3.42	1.07	0.64
FeO*	86.42	79.88	73.56	77.71
FeO	36.21	38.23	48.71	40.61
Fe_2O_3	55.84	46.31	27.64	41.26
MnO	0.11	0.21	0.49	0.20
MgO	0.05	0.01	0.36	0.03
ZnO	0.00	0.00	0.00	0.00
Total Fe0*	95.35	93.29	97.48	92.00
Total	100.98	97.95	100.27	96.16

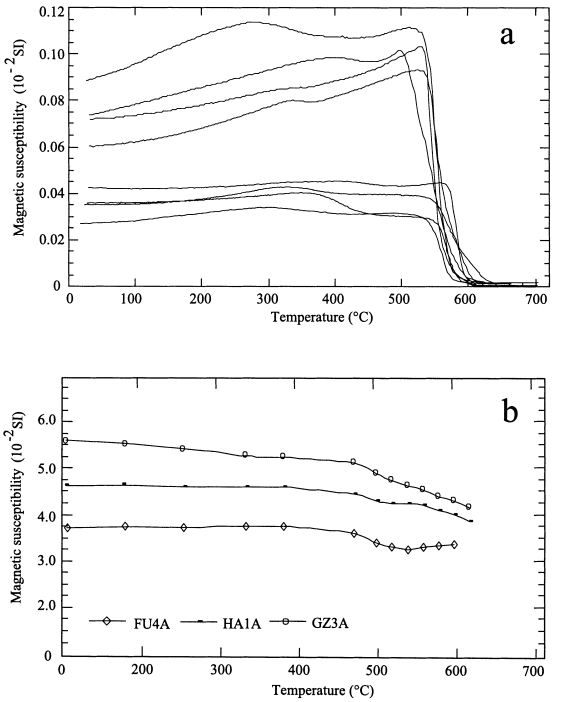


Fig. 4. (a) Magnetic susceptibility variation versus temperature for some dykes of French Guyana showing a susceptibility drop between 550 and 575°C. (b) Bulk susceptibility variation during thermal demagnetization from selected samples (dykes FU4A, HA1A and GZ3A).

Petrographic observations with reflected light microscopy show large automorphous grains of titanomagnetite (Fig. 3a). Scanning electron microscopy (SEM) observations present more detailed information on the structure of the large titanomagnetite grains. Fig. 3b,c show two secondary electrons images of a 0.5 mm titanomagnetite grain at two scales. A complex fractal structure of ilmenite lamella exsolutions has been observed in the titanomagnetite crystal body (Fig. 3c). This kind of structure is well known in basaltic rocks and results from the exsolution

of ilmenite during magmatic cooling between 550 and 515°C [21]. No later destabilization was observed in titanomagnetite. The microprobe analyses on titanomagnetite are listed in Table 1. Titanium oxide concentration ranges from 4 to 20 percent. Thermomagnetic experiments confirm that titanomagnetite is the main magnetic carrier (Fig. 4a). Indeed, Fig. 4a shows a sharp drop of the low field magnetic susceptibility between 550 and 575°C. Bulk susceptibility measurements made after each heating step show only minor variations which indicate that the magnetic mineralogical transformations during heating are insignificant (Fig. 4b).

In short, petrographic and magnetic mineralogical analyses indicate that the remanent magnetization in the dolerites is carried solely by minerals of the titanomagnetite family.

Table 2
AMS data

Sites	F	L	P	T	K_1				K_2				K_3				Fabric type
					D	I	α_{95min} (°)	α_{95max} (°)	D	I	α_{95min} (°)	α_{95max} (°)	D	I	α_{95min} (°)	α_{95max} (°)	
FU	1.010	1.007	1.017	0.152	211	13	6	8	281	36	10	11	105	53	7	10	R
GR	1.010	1.019	1.029	−0.301	38	1	5	6	300	79	7	24	128	11	3	24	R
GT	1.005	1.013	1.018	−0.488	220	33	8	12	341	39	9	30	104	34	7	28	R
GU	1.023	1.017	1.040	0.141	175	7	6	8	68	67	5	9	267	22	5	7	N
GV	1.065	1.017	1.082	0.577	186	3	3	9	283	67	1	9	95	22	3	4	N
GW	1.032	1.008	1.040	0.610	267	68	9	40	19	8	14	40	112	20	8	13	N
GZ	1.019	1.013	1.032	0.188	199	0	4	7	108	54	4	7	289	36	5	6	R
HA	1.005	1.005	1.011	0.100	190	20	12	14	303	41	11	13	81	51	7	15	N
O1	1.011	1.008	1.019	0.163	340	1	3	5	239	82	5	6	70	8	4	6	N
O3	1.006	1.014	1.020	−0.421													N
O4	1.004	1.014	1.018	−0.542	52	61	11	13	213	24	8	27	320	8	7	28	N
O5	1.023	1.011	1.034	0.366	337	24	5	13	240	16	6	8	120	60	4	7	N
O7	1.047	1.016	1.063	0.482	157	18	5	13	256	25	5	13	49	54	3	6	I
O9	1.031	1.013	1.045	0.403	168	2	1	14	263	38	2	4	189	60	1	13	N
10	1.021	1.014	1.036	0.239	334	23	9	10	198	59	3	12	70	20	6	9	N
15	1.011	1.012	1.023	−0.047	252	21	6	8	30	62	13	28	353	15	5	13	R
17	1.038	1.018	1.057	0.345	154	1	5	9	251	80	4	28	64	10	5	7	N
23	1.037	1.011	1.049	0.549	345	13	7	10	137	76	2	8	254	7	2	9	N
PD	1.039	1.002	1.041	0.924	157	2	5	13	249	40	4	13	64	49	3	5	N
PI	1.055	1.013	1.068	0.601	351	12	4	13	183	78	4	13	82	2	3	6	N
OU	1.005	1.005	1.009	0.012													O
GosD1	1.011	1.014	1.025	0.100	252	13	4	7	345	11	6	11	115	73	4	6	R
BouD6	1.020	1.011	1.032	−0.295	46	1	4	13	136	1	8	18	275	88	4	6	R
CY	1.003	1.015	1.018	0.667	184	14	6	19	73	53	5	19	316	55	3	12	N
CQ	1.011	1.038	1.049	0.560	187	21	4	25	289	29	14	26	66	53	2	15	N
PolD4	1.018	1.007	1.024	−0.376	44	17	4	33	309	14	9	14	171	72	12	13	R
polD5	1.004	1.010	1.015	0.401													O
AN	1.010	1.020	1.030	0.292	145	12	6	15	240	57	8	19	59	18	7	12	N
AQ	1.008	1.043	1.051	0.683	159	60	1	23	9	31	4	23	259	16	3	4	N
BN	1.035	1.027	1.064	−0.123	136	17	4	16	295	71	4	16	232	7	4	17	N
BO	1.019	1.018	1.037	−0.050	348	0	7	19	253	84	14	22	78	6	9	18	N
BP	1.028	1.053	1.085	0.186	159	3	3	17	253	39	10	37	80	50	6	30	N
BQ	1.004	1.007	1.011	0.297													O
BR	1.003	1.036	1.039	0.859	158	5	12	24	60	57	8	25	252	33	10	14	N

K_1, K_2 and K_3 are the maximum, intermediate and minimum susceptibility intensities; F, mean foliation (K_2/K_3); L, mean lineation (K_1/K_2); P, mean anisotropy degree ($P=K_1/K_3$); T, Jelinek's parameter [$T=(\ln(K_2/K_3)-\ln(K_1/K_2)-\ln(K_2/K_3)]$; D, declination; I, inclination; α_{95min} (°) and α_{95max} (°): bimodal statistic data [22]. Fabric types: N, 'Normal'; R, 'Reversal'; I, 'Intermediate'; O, 'Other' (see text).

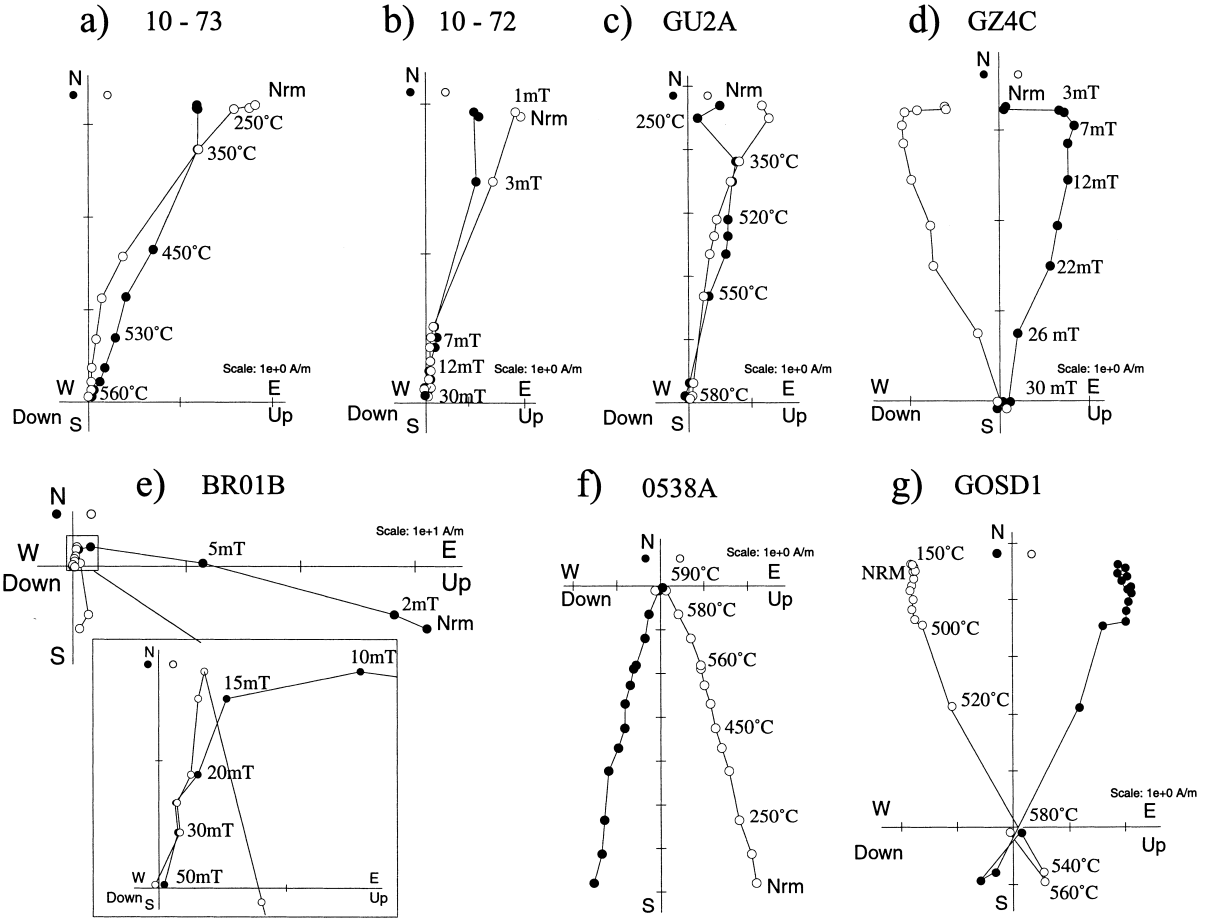


Fig. 5. Representative orthogonal vector diagrams of progressive thermal and AF demagnetization. The closed (open) symbol refer to the horizontal (vertical) plane. (a)–(d) Normal polarity directions, (e) shows a specimen affected by partial lightning, (f) and (g) show reversed directions.

3.2. AMS

AMS measurements were performed on each specimen using a KLY-3S (AGICO, Geofysica) spinner kappabridge. The AMS results are listed in Table 2 and their tectonic interpretations are discussed in Section 4. The statistic precisions were calculated with bimodal statistics [22] and listed in Table 2. The anisotropy degree (P) is defined by K_1/K_3 where K_1, K_3 are the maximum and minimum susceptibility intensity axes respectively. It is relatively low, varying from 1.009 to 1.082 with an average of 1.03 (3% anisotropy) which is typical order found in dykes [23]. Ellipsoid shape parameter T, expressed by $T=(\ln F-\ln L)/(\ln L-\ln F)$ where $F=K_2/K_3$ and $L=K_1/K_2$ shows that it is predominantly (25 dykes) oblate ($T>0$).

3.3. Paleomagnetic analyses

Magnetic remanence was measured on a JR5 spinner magnetometer (AGICO, Geofysica). Natural remanent magnetization (NRM) measurements show a bimodal distribution, ranging from 1 to 10 A/m for one group (75%) and 70–100 A/m for other. The group with high intensity values probably suffered an isothermal magnetization due to lightning.

A pilot study was carried out on a few speci-

Table 3
Summary of sampling location and paleomagnetic results with corresponding ^{40}Ar–^{39}Ar data from French Guyana and Amapa studied dykes

Location	Localization			^{40}Ar–^{39}Ar Ages After [3] mini plateau ages (^{39}Ar released)	Mean direction						Polarity	VGP		
	Site	Lat. (°N)	Long. (°E)	(Ma)	*n*/*N*	*D* (°)	*I* (°)	*k*	α_{95} (°)		λ (°N)	ϕ (°E)	dp (°)	dm (°)
Cayenne Island	BouD6	4.9	307.8		6/7	2	−22.3	60.8	8	N	73.4	120.9	4.5	8.5
	CQ	4.9	307.8		4/5	325.1	7.8	14.2	21	N	55.4	212.3	10.5	21[a]
	PolD4	4.9	307.8		5/6	3.4	−3.5	294.4	4.5	N	82.5	100.8	2.3	4.5
	PolD5	4.9	307.8		5/5	0.4	2.1	135.5	6.6	N	86.1	121.8	3.3	6.6
Approuague River	AN	4.2	307.7	198.3 ± 2.0 (22%)	8/8	355.4	−18.4	10	18.5	N	75.5	146.3	10	19.2[a]
	BN	4.2	307.7		7/10	350.4	−16.3	10.2	19.9	N	74.3	165.8	10.6	20.6[a]
	BQ-BR	4.1	307.5		4/14	352.1	−13.7	9.4	33.3	N	76.4	163.1	17.4	34[a]
	BO	4.2	307.9		7/7	5.1	−4.9	37.9	9.9	N	81.6	90.2	5	9.9
	BP	4.2	307.7		6/6	356.1	5.7	167.2	5.2	N	85.9	199.3	2.6	5.2
	AQ	4.2	307.9		8/8	2.2	−7	49.8	7.9	N	81.9	111.9	4	7.9
Maroni River	15	5	305.6	196.0 ± 1.7 (40%)	5/6	9.5	−8.3	75.7	8.9	N	76.8	79.5	4.5	9
	09	5.2	305.7			Random[a]								
	10	5.2	305.7		4/6	26.6	−12.7	55.5	12.4	N	62.1	58.9	6.4	12.6
Mana River	17	5.2	306.4	196.0 ± 5.7	6/7	4.3	1.3	50.3	9.5	N	83.7	82.9	4.8	9.5
	23	5.3	306.4		3/7	3.9	−10.9	17.9	30	N	78.6	106.3	15.4	30.4[a]
North Oyapok River	GW	3.3	308		5/5	20	−23.5	47.6	11.2	N	64.8	75.3	6.2	11.7
	GR	3.2	307.8		5/7	29.7	−36.7	15	20.4	N	53	78.1	13.4	23.8[a]
	FU	3.7	308		6/10	7.9	4.4	92.9	7	N	82	48.5	3.5	7
	GU	3.4	307.9		9/10	12.4	−4.5	228.2	3.4	N	76.4	62.3	1.7	3.4
	GV	3.4	307.9		4/6	1	−1.5	44.2	14	N	85.7	114.3	7	14
	GT	3.4	307.9		5/6	348.1	12.3	66.8	9.4	N	77.8	231.7	4.9	9.6
South Oyapok River	OU	2.3	307.5		4/6	5.7	−20.6	65.8	11.4	N	76	103.9	6.3	12
	PD	2.2	307		5/6	2.7	−13.6	359.7	4	N	80.5	110.6	2.1	4.1
	PI	2.3	307		5/6	4.3	−5.4	49.8	11	N	83	86.3	5.5	11
Amapa (Cassiporé	03, 04	1.5	309.1		8/12	357.3	7.8	67	6.8	N	86.4	261	3.4	6.8
	01	2.5	308.3			Random[a]								
	mean direction Pole B			**18/26**			***k* = 69.8**	**A_{95} = 4.2°**		**81.6**	**89.1**			
Cayenne Island	GosD1	4.9	307.8		22/32	22.3	19.8	55.6	3.4	N/R	67.3	23	1.9	3.6
	GosD2	4.9	308.8		6/8	17.3	17.5	37.9	9.2	N	72.4	23.6	4.9	9.5
	GosF1	4.9	307.8		6/4	16.4	23.8	48.9	13.3	N	72.1	11.8	7.6	14.2
	CY	4.9	307.8		5/5	19.1	18.7	44.5	11.6	N	74.1	22.6	6.3	12.1
Conté River	05	4.5	307.5		6/6	195.5	−16.4	89.1	7.1	R	79.8	16.6	3.8	7.3
	07	4.5	307.5	192.3 ± 2.0 (49%)	5/6	9.5	16	60.2	9.9	N	79	15.8	5.2	10
	mean direction Pole A			**8/8**			***k* = 288.8**	***A_{95}* = 3.4°**		**73.2**	**15.3**			
	mean direction M			**26/34**			***k* = 50.7**	***A_{95}* = 4.0°**		**81.2**	**55.1**			

[a]Sites discarded for mean directions and poles. Age errors are indicated at the 1σ level [3].

mens using both thermal and alternating field (AF) demagnetization techniques with a Pyrox furnace and an automated three axes tumbler AF demagnetizer (LDA-3, AGICO, Geofysica), respectively. About 12 (AF) to 16 (thermal) demagnetization steps were used. Both techniques provided similar results (Fig. 5a,b) as defined by principal component analysis [24]. After removal of a low coercivity or low unblocking temperature component with random directions, the intensity linearly decays up to maximum applied fields of 30–40 mT or temperatures ranging from 530 to 580°C (Fig. 5a–d). For samples possessing high NRM intensities, the AF technique more efficiently isolates the characteristic component after removing the spurious overprint (Fig. 5e). We therefore used AF demagnetization on all other specimens. The magnetic directions are mainly of normal polarity although reversed polarity directions were identified in two sites (Fig. 5f,g). One Site (05) has only reversed polarity directions while the other (GOS D1) a 20 m wide dyke presents few reversed polarity directions only in the slow cooling coarse grained central part.

The mean direction for each paleomagnetic site, given in Table 3, have been computed by Fisher statistics [25]. They are slightly but systematically different from the present Earth field (declination = 342°, inclination = 18.5°) and the geocentric axial dipole field (inclination = 3–10°). The Fisher (k) parameter is usually greater than 10. Directions isolated from the highly magnetized specimens usually led to a poorly defined mean direction with a low k and a large α_{95} (Table 3). The sites which have a low precision parameter value and a α_{95} greater than 15° (Table 3) were not taken for mean direction and mean virtual geomagnetic pole (VGP) calculations.

4. Discussion

4.1. Implication of AMS results on dyke emplacement

Based on numerous field observations and AMS studies of characteristics of eigenvector orientations, many authors (e.g. [10,23]) suggest that magmatic fabrics in dykes can be classified into four general types: N (Normal), R (Reversal), I (Intermediate) and O (Other). Type N is characterized by the clustering of K_1 and K_2 axes on the dyke plane, whereas K_3 is nearly perpendicular to it. Type I is defined by K_1 and K_3 axes clustering close to the dyke plane, whereas K_2 axis is perpendicular to this plane. Type R is characterized by K_2 and K_3 axes close to the dyke plane and K_1 is perpendicular to that plane. AMS axes which do not show any preferential orientation belong to Type O.

In our dykes, the analysis at the individual-dyke scale defines four fabrics (Table 2 and Fig. 6): N (21 dykes, 62%), R (8 dykes, 24%) I (1 dyke, 2%) and O (4 dykes, 12%).

The K_1 orientation of Type N (Figs. 6 and 7) is parallel to the NNW–SSE strike of the dykes. The N fabric type was interpreted as a flow fabric by many researchers (e.g. [23,26,27]) who compared field evidence and AMS data. Moreover, the K_1 inclination can be used to infer the distance between the fractures and the magma source [23,28]. The subhorizontal K_1 inclinations could correspond to dykes fed horizontally by a distant feeding source. In French Guyana dolerite, all N fabric K_1 axes have weak to horizontal inclinations (Figs. 6 and 7), suggesting that French Guyana dolerites are, firstly far from their feeding source (>500 km) and, secondly fed horizontally from these feeding source and/or magma storage zone. We also notice that the weak K_1 axes generally dip towards the south (Fig. 7). However, they are almost vertical at one site from Amapa (Site 04; Fig. 6), indicating that this dyke may probably be closer to the magma source with respect to the French Guyana dykes. If the French Guyana and Amapa dykes belong to the same generation and have same magma source, the source could be located to the SE. This conclusion contradicts some workers [29,30], who suggest that the source of the French Guyana dykes was located to the NNW, based on their geographical location and the preferred circum Central Atlantic Ocean dyke orientation.

The reversal fabric was found in eight dykes. It was generally attributed to secondary processes such as hydrothermalism, post-emplacement mod-

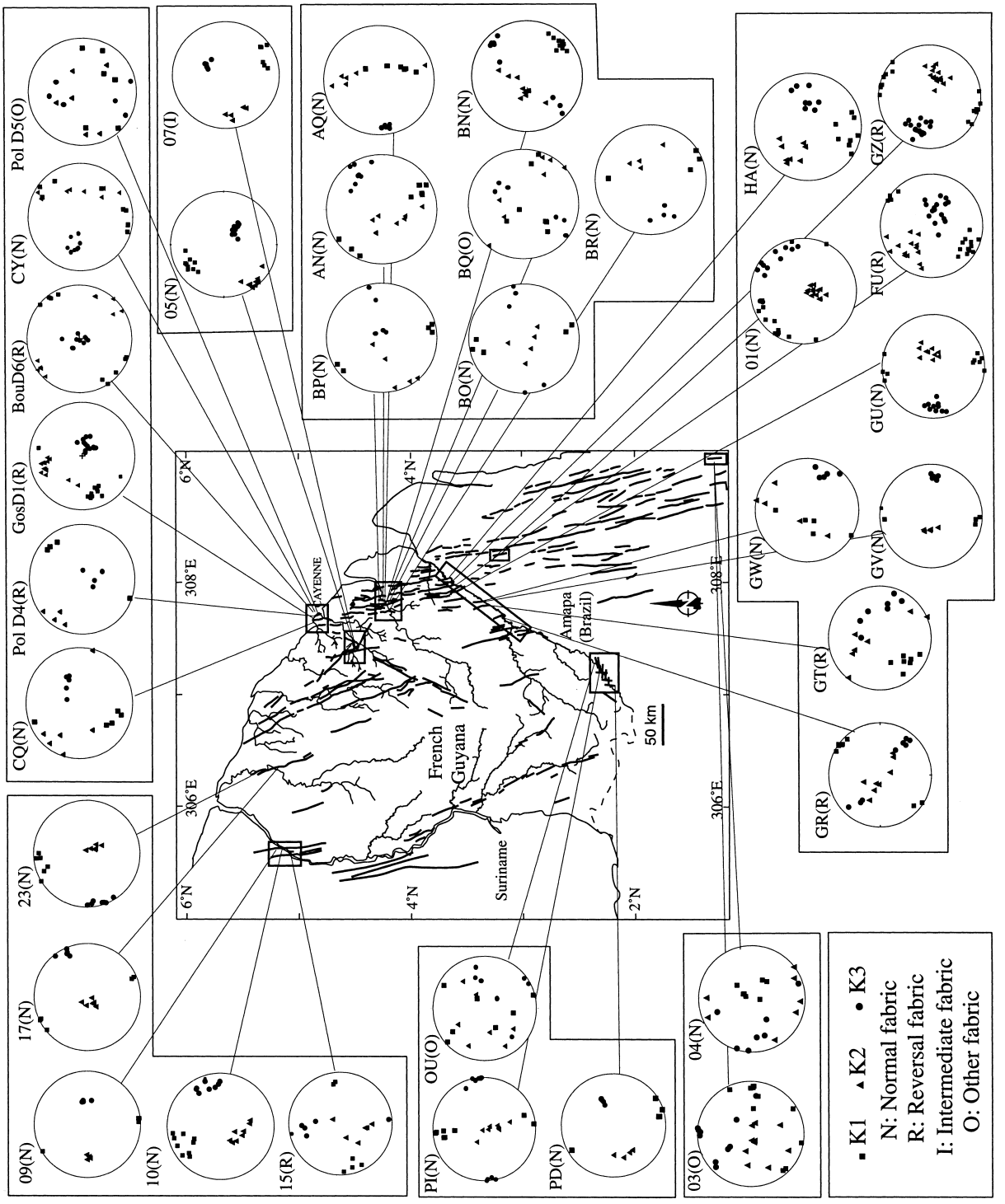


Fig. 6. AMS directions (K_1, K_2 and K_3) from the sampled dolerites of French Guyana. Magmatic fabrics are highlighted as follows: N, 'Normal'; R, 'Reversal'; I, 'Intermediate'; O, 'Other'.

ification or due to the presence of SD (single domain) grains in rocks [23]. If the analysis in thin sections does not show evidence of hydrothermalism or solid state deformation, shearing deformation has been locally shown by field observations, in contact zone between dykes of Type R and their host rocks (e.g. GOSD1, BOUD6). We suggest that the R fabric could be related to local shearing deformation.

The 'Intermediate' fabric (I Type) was recognized in one dyke (Site 07; Fig. 6). Such fabric was found in other swarm [23,31]. It could be explained by the presence of fine-grained (titanomagnetite) [23] or due to the vertical compacting of a static magma column [31]. In this study, petrographic investigation has not permitted to find fine magnetic grain size in this rocks (see Section 3.1), so the I fabric may be explained by the compression tending to force the material along the dyke direction during dyke emplacement [31].

4.2. Implication of paleomagnetic results

All site-mean directions and poles are listed in Table 3. Due to the large sampling area, all site-mean directions have been transferred to corresponding VGPs before calculating mean poles. The site-mean VGPs are systematically distinct from the present geomagnetic field. Cores collected parallel to the dyke width show consistent directions with generally satisfactory statistic parameters (Table 3). The majority of sites present normal polarities and only two sites show a reversed one (Site 05, GosD1). We also note that the strong weathering which affects this subequatorial area has not yet induced any change in the magnetization of this collection. Indeed, it has been recently shown [32], in the Cayenne area, that independently of the primary magnetic carrier, hematite becomes the main magnetic carrier due to lateritization and no magnetite remains, which is not the case in the present study. Based on the above observations, we can conclude that the magnetic remanence is probably of primary origin representing an Early Jurassic magnetization.

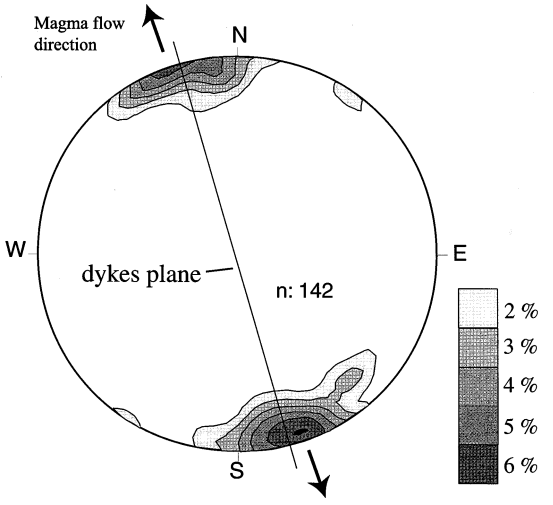


Fig. 7. Density diagram (1% area contours, equal area) of the magnetic lineation (K_1 direction) from all samples belonging to the magmatic fabric of Type N.

Eight of 34 sampled sites show random or scattered directions due to a partial or to a total overprint attributed to lightning. They were rejected from the mean calculation. From the remaining 26 sites, we calculated an Early Jurassic paleomagnetic pole for French Guyana (Table 3; Fig. 8a; Pole M) with the following coordinates: $\lambda = 81.2°N$, $\phi = 55.1°E$, $k = 50.7$ and $A_{95} = 4.0°$.

A closer look at the VGP distribution shows that two populations could be identified in our data set (Fig. 8a). The disparity between these two populations is predominantly on the polar longitude. Therefore, two sub-mean poles are computed: Pole A: $\lambda = 73.2°N$, $\phi = 15.3°E$, $k = 288.8$, $A_{95} = 3.4°$ and $N = 8$; Pole B: $\lambda = 81.6°N$, $\phi = 89.1°E$, $k = 69.8$, $A_{95} = 4.2°$ and $N = 18$. These two poles are statistically distinguished suggesting two distinct magmatic intrusive events (Fig. 8a), although secular variation of the geomagnetic field could also be responsible for part of the data scattering. The only normal polarity evidenced in the majority of dykes may indicate that duration of magmatic intrusion for each group could be very short as reversal rate is relatively high for this Triassic–Jurassic period [33]. Moreover, the two groups do not show the same distribution style, the VGPs from Group B are more dispersed than those from Group A, which may indicate that the intrusion of Group B may have a longer duration. No clear relation-

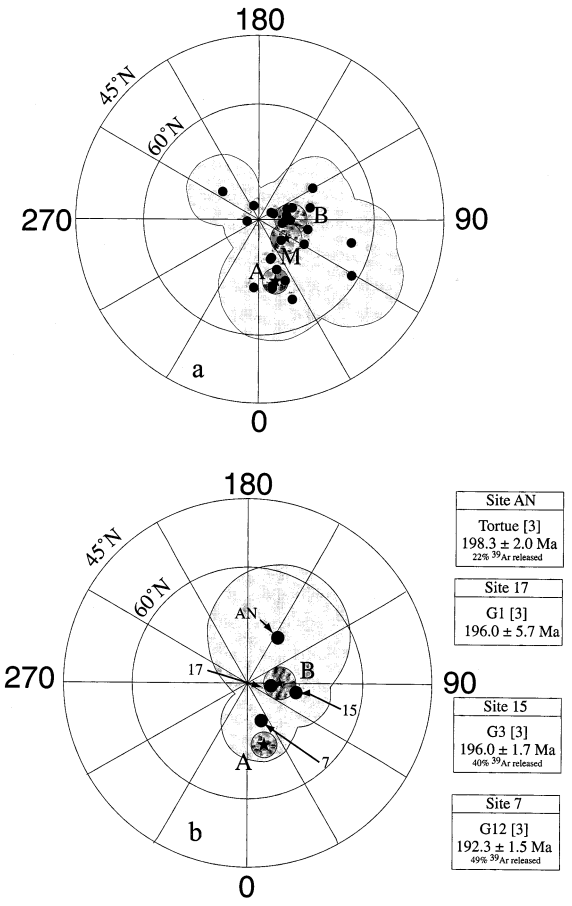


Fig. 8. (a) Equal area projection of the 26 VGP from the French Guyana dolerites with the total mean pole (M) and the proposed pole A and pole B. (b) VGPs from the dated dykes of [3] with our proposed A and B poles. Age errors are given with 1σ level.

ship was observed between ASM types and the two VGP groups.

Four of our sampled dykes were recently dated by ^{40}Ar–^{39}Ar [3]. Sample G1 (Site 17) corresponds to a single grain amphibole age of 196.0 ± 5.7 Ma. According to the authors [3] the large error bars are due to the low K content and small grain size. The ages obtained from samples Tortue corresponding to the paleomagnetic site: AN, G3 (15), and G12 (7) were determined from plagioclase. All spectra are characterized by variable excess argon and younger K-rich alteration phases. Nevertheless, Deckart et al. [3] indicated that between 770 and 1090°C the age spectra display a relatively flat region corresponding to 20–49% of total out gassed ^{39}Ar and show three to five concordant successive apparent ages. They also argued that these 'mini-plateau' ages correspond to a minimum contribution of excess argon and/or radiogenic ^{40}Ar from K-rich alteration phases. Furthermore, they presented a bimodal age distribution between 196 and 189 Ma with the 'mini-plateaus' probability diagram of bulk plagioclase of French Guyana dolerites (see figure 6 of [3]). In Fig. 8b, we represent VGPs location sites AN, 15 and 7 and corresponding Tortue, G3, G12 'mini plateau' ages. The two older indistinguishable ages (198.3 ± 2.0 Ma and 196.0 ± 1.7 Ma) sites AN and 15 lie on between 150 and 90°E longitude. The younger Site 7 (192.3 ± 1.5 Ma) is between 0 and 30°E.

According to VGP distribution and geochronological data [3], we propose that the two VGP groups could correspond to two distinct magmatic pulses occurring in a short time period. Moreover, all eight sites which define the pole A come from the northeastern part of French Guyana (Cayenne Island, Conté and North Oyapok), while the sites for the pole B cover all sampling area. As proposed by [2] in Brazilian CAMP, the youngest magmatic event seems to be restricted closer to the future rift margin.

5. Conclusions

AMS and paleomagnetic studies on recently dated dykes from French Guyana bring new and important insight on the Early Jurassic circum Atlantic volcanic activity which occurred prior to the opening Central Atlantic Ocean. AMS can be segregated into four fabric types according to the eigenvector orientations with respect to the dyke plane. Type O shows no preferential orientation of the principal axes of anisotropy. The three others (N,R,I) present well defined K_1, K_2 and K_3 directions. Type N referred as 'Normal' was found in 21 dykes and defined by K_1–K_2 clustering close to the dyke plane and K_3 is nearly perpendicular to it. This fabric was interpreted as due to the magma flow. Analysis of K_1 inclination permitted to infer that the dykes of French

Guyana were fed horizontally from a distant magma source. Type R (8 dykes), is characterized by a K_2-K_3 plane close to the dyke plane and K_1 perpendicular to dyke orientation. The origin of this magmatic fabric is due to possible local shear stress. Type I, defined by K_1–K_3 axes close to the dyke plane and K_2 perpendicular to this plane, was interpreted as due to vertical compaction of the magma column.

The paleomagnetic results from 26 out of 34 sites in French Guyana suggests the existence of two VGP populations and indicates that two short volcanic events took probably place in the French Guyana during the Early Jurassic. The geographical position of the dykes from the two VGP groups indicated a possible migration of the magmatism toward the future rifting zone in French Guyana. To validate these hypotheses, more coupled paleomagnetic and geochronological studies have to be undertaken on similar rocks from the Amazonian Shield and CAMP.

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